Discrete Variational Calculus on the Hamilton Principle

$$\mathscr{A}^\star[q,p]=\int_{t_0}^{t_N}\big(\dot q p - L^\star(q,p)\big)\mathrm{d}t$$

Define a discrete Hamiltonian

$$\mathscr{A}^\star_{\mathrm{d}}=\sum_{n=0}^{N-1}\left(\frac{q_{n+1}-q_n}{\Delta t}p_{n+1/2}-L^\star_{\mathrm{d}}(q,p)\right)$$

$$L^\alpha_{\mathrm{d}}=p_{n+(1-\alpha)}\frac{q_{n+1}-q_n}{\Delta t}-L^\star_{\mathrm{d}}\big(q_{n+\alpha},p_{n+(1-\alpha)}\big)$$

$$L^{\alpha=0}_{\mathrm{d}}$$

$$L^{\alpha=1}_{\mathrm{d}}$$

$$L_{\mathrm{d}}=\frac{\Delta t}{2}\left(L^{\alpha=0}_{\mathrm{d}}\Big|_{t\in[t_n,t_{n+1/2}]}+L^{\alpha=1}_{\mathrm{d}}\Big|_{t\in[t_{n+1/2},t_{n+1}]}\right)$$

Discretisation of the Hamilton Equations

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}q\\p\end{pmatrix}=\begin{pmatrix}0&1\\-1&0\end{pmatrix}\begin{pmatrix}\partial L^\star/\partial q\\\partial L^\star/\partial p\end{pmatrix}$$

Choose a scheme that preserves the structure $\dot z=\tilde{\mathbb{J}}\nabla L^\star$

$$\frac{z_{n+1}-z_n}{\Delta t}=\tilde{\mathbb{J}}\nabla L^\star(z_{n+\alpha},z_n)$$

$$\Phi^\alpha:(q_n,p_n)\mapsto\begin{cases}q_{n+1}=q_n+\Delta t\dfrac{\partial L^\star}{\partial p}(q_{n+\alpha},p_{n+(1-\alpha)})\\ p_{n+1}=p_n-\Delta t\dfrac{\partial L^\star}{\partial q}(q_{n+\alpha},p_{n+(1-\alpha)})\end{cases}$$

$$\Phi^{\alpha=0}$$

$$\Phi^{\alpha=1}$$

$$\left(\Phi^{\alpha=0}\Big|_{t\in[t_n,t_{n+1/2}]}+\Phi^{\alpha=1}\Big|_{t\in[t_{n+1/2},t_{n+1}]}\right)$$

Discrete Variational Calculus on the Poisson Bracket

$$\frac{\mathrm{d}F}{\mathrm{d}t}=\begin{pmatrix}\partial F/\partial q\\\partial F/\partial p\end{pmatrix}^{\mathsf{T}}\begin{pmatrix}0&1\\-1&0\end{pmatrix}\begin{pmatrix}\partial L^\star/\partial q\\\partial L^\star/\partial p\end{pmatrix}\text{or}\quad\frac{\mathrm{d}F}{\mathrm{d}t}=\{F,L^\star\}$$

$$\frac{\partial F}{\partial z}\left(\dot z-\tilde{\mathbb{J}}\nabla L^\star\right)$$

$$\sum_{n=0}^{N-1}\int_{t_n}^{t_{n+1}}\frac{\partial F}{\partial z}\dot z\,\mathrm{d}t=\sum_{n=0}^{N-1}\int_{t_n}^{t_{n+1}}\{F,L^\star\}\,\mathrm{d}t$$

Choose the appropriate interpolation

$$\Delta t\frac{\partial F}{\partial z}\frac{z_{n+1}-z_n}{\Delta t}=\Delta t\{F,L^\star\}$$

$$\begin{pmatrix}\dfrac{\partial f}{\partial q}(q_{n+\alpha},p_{n+(1-\alpha)})\\\dfrac{\partial f}{\partial p}(q_{n+\alpha},p_{n+(1-\alpha)})\end{pmatrix}\begin{pmatrix}\dfrac{q_{n+1}-q_n}{\Delta t}\\\dfrac{p_{n+1}-p_n}{\Delta t}\end{pmatrix}=\begin{pmatrix}\dfrac{\partial f}{\partial q}(q_{n+\alpha},p_{n+(1-\alpha)})\\\dfrac{\partial f}{\partial p}(q_{n+\alpha},p_{n+(1-\alpha)})\end{pmatrix}\begin{pmatrix}0&1\\-1&0\end{pmatrix}\begin{pmatrix}\dfrac{\partial l^\star}{\partial q}(q_{n+\alpha},p_{n+(1-\alpha)})\\\dfrac{\partial l^\star}{\partial p}(q_{n+\alpha},p_{n+(1-\alpha)})\end{pmatrix}$$

$$F^{\alpha=0}_{\mathrm{d}}$$

$$F^{\alpha=1}_{\mathrm{d}}$$

$$\int_{t_n}^{t_{n+1}}F_{q,p}\,\mathrm{d}t=F^{\alpha=0}_{\mathrm{d}}\Big|_{t\in[t_n,t_{n+1/2}]}+F^{\alpha=1}_{\mathrm{d}}\Big|_{t\in[t_{n+1/2},t_{n+1}]}$$

α symplectic Euler scheme

$$\begin{cases}\dfrac{q_{n+1}-q_n}{\Delta t}=\dfrac{\partial l^\star}{\partial p}(q_{n+\alpha},p_{n+(1-\alpha)})\\\dfrac{p_{n+1}-p_n}{\Delta t}=-\dfrac{\partial l^\star}{\partial q}(q_{n+\alpha},p_{n+(1-\alpha)})\end{cases}$$

Implicit symplectic Euler

$$\begin{cases}\dfrac{q_{n+1}-q_n}{\Delta t}=\dfrac{\partial l^\star}{\partial p}(q_n,p_{n+1})\\\dfrac{p_{n+1}-p_n}{\Delta t}=-\dfrac{\partial l^\star}{\partial q}(q_n,p_{n+1})\end{cases}$$

Explicit symplectic Euler

$$\begin{cases}\dfrac{q_{n+1}-q_n}{\Delta t}=\dfrac{\partial l^\star}{\partial p}(q_{n+1},p_n)\\\dfrac{p_{n+1}-p_n}{\Delta t}=-\dfrac{\partial l^\star}{\partial q}(q_{n+1},p_n)\end{cases}$$

Central difference

$$\begin{cases}q_{n+1}=q_n+\dfrac{\Delta t}{2}\left(\dfrac{\partial l^\star}{\partial p}(q_{n+1},p_{n+1/2})+\dfrac{\partial l^\star}{\partial p}(q_n,p_{n+1/2})\right)\\p_{n+3/2}=p_{n+1/2}-\dfrac{\Delta t}{2}\left(\dfrac{\partial l^\star}{\partial q}(q_{n+1},p_{n+1/2})+\dfrac{\partial l^\star}{\partial q}(q_{n+1},p_{n+3/2})\right)\end{cases}$$